Effect of Potassium Promotion on Bimetallic Iron-Cobalt Catalysts for Hydrogen Release from Ammonia

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Introduction

Among the numerous catalysts developed for ammonia decomposition, the Ru catalysts exhibited superior activity due to its moderate/suitable nitrogen binding energy, which is an indicator for evaluating the reactivity of transition metals in ammonia decomposition.[1] However, the scarcity and high cost of Ru limits its large-scale industrial application. Recently, we have developed an inexpensive and highly active bimetallic iron-based catalysts by alloying with cobalt, resulting in a moderate metal-nitrogen binding energy.[2] However, the bimetallic FeCo catalyst still shows room for improvement in its activity for ammonia decomposition. Generally, by introducing electronic promoters such as potassium (K) into the catalytic system, the desorption of nitrogen (N_2) from the catalyst surface is facilitated, further enhancing the catalytic activity.[3] Here, we will introduce K into the FeCo catalyst to investigate the effect of K promotion on the activity for ammonia decomposition.

Experimental Section

$Mg(Fe_{0.5}Co_{0.5})_2O_4$ used in this study was synthesized by co-precipitation of layered double hydroxide (LDH) precursors in an automated laboratory reactor system (Optimax, Mettler Toledo). The K-FeCo/MgO catalysts were prepared by loading the desired amount of $KHCO_3$ solution onto the LDHs via incipient wetness impregnation. The K loaded precursors were subsequently calcined and further isothermally reduced in H_2. Kinetic measurements for the NH_3 decomposition were carried out in a fixed-bed flow reactor in a gas stream of 100% NH_3 at a flow of 60 sccm. The outlets were analyzed by micro-GC (Inficon). Structural analysis of the K-FeCo/MgO catalysts was carried out by using XRD, XPS and TEM. NH_3-TPD as well as TPSR measurements were performed in a BelCat II reactor system (Microtrac).

Results and Discussion

Potassium significantly improved the catalytic activity of FeCo/MgO catalysts in ammonia decomposition (Fig. 1). The most active samples with 0.15 wt.%- and 0.31 wt.%-K showed an around 3-fold increase in hydrogen formation rate at 500 °C

compared to unpromoted FeCo/MgO. Higher potassium loading resulted in a gradual decrease in the rate, which may be due to the agglomeration of K species on the active sites. Compared to FeCo/MgO, K-FeCo/MgO catalysts with low K loadings (0.08–0.76wt.%) showed a slightly smaller NH_3 reaction order (~0.45). While K-FeCo/MgO with high K loadings (1–8 wt.%) exhibited a clear increase of the NH_3 reaction order to ~1.04 (Fig. 2). This indicates that the participation of ammonia in the reaction becomes more important on the FeCo catalysts with high K loading, assuming that the rate-limiting step under these conditions is the NH_X dehydrogenation. The TPSR profiles obtained during NH_3-TPD measurements showed a shift of the N_2 desorption to lower temperatures in the presence of low K concentrations, proving the electronic promoting effect of potassium, which facilitates N_2 desorption. Further characterizations by TEM and XPS will be reported in our contribution.

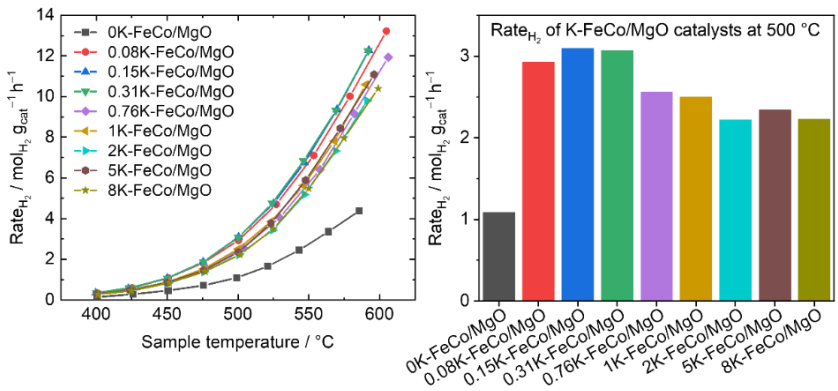

Fig. 1. Hydrogen production rate of K-FeCo/MgO catalysts with different K loadings.

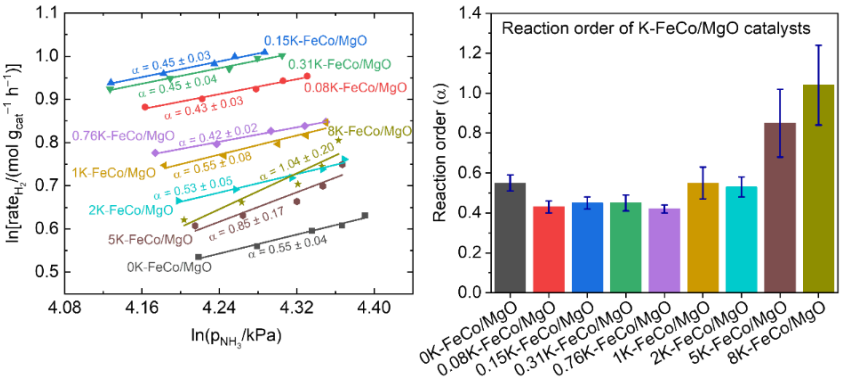

Fig. 2. The ammonia reaction order of K-FeCo/MgO catalysts at 500°C in ammonia decomposition.

References

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